

Lab Manual for Digital Logic and Design

LAB-02 **Verification of basic identities of Boolean algebra**

EXPERIMENT NO. : 02

Verification of basic identities of Boolean algebra:

- Identity Law
- Null (or Dominance) Law
- Idempotent Law
- Inverse Law
- Double Complement Law
- Associative Property
- Distributive Property

Objective:

To understand basic identities of Boolean algebra

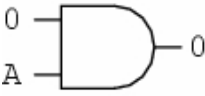
Apparatus:

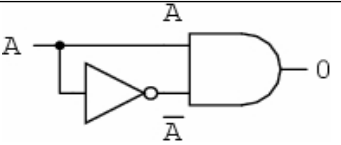
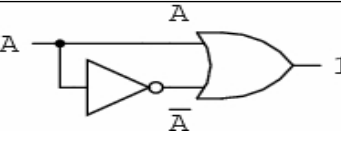
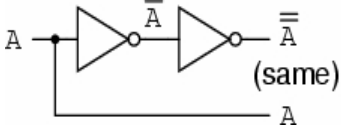
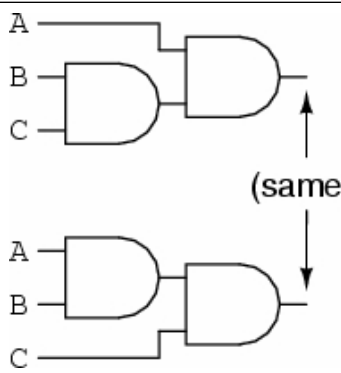
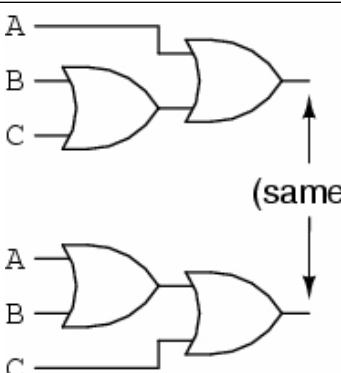
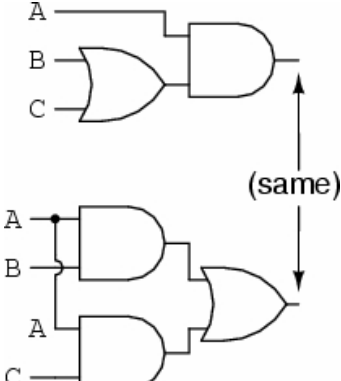
- Logic trainer/MultiSim
- Connecting wires
- 14 pin ICs(7408,7400,7432,7402,7486,74266,7404)
- Power supply

Procedure:

- Connect circuits for each law separately
- Connect the inputs to the input switches provided in the trainer board.
- Connect the outputs to the switches of output LEDs.
- Apply various combinations of inputs as shown in the truth table and observe the Conditions of LEDs.

Basic Identities of Boolean Algebra

Identity Name	AND Form	OR Form
Identity Law	 $1A=A$	 $A+0=A$
Null (or Dominance) Law	 $0A=0$	 $A+1=1$
Idempotent Law	 $AA=A$	 $A+A=A$

Inverse Law	 $AA' = 0$	 $A + A' = 1$
Double Complement Law	 $A'' = A$	
Associative Property	 $A(BC) = (AB)C$	 $A + (B + C) = (A + B) + C$
Distributive Property	 $A(B + C) = AB + AC$	

Simulations:

Identity Law (AND or OR Form)

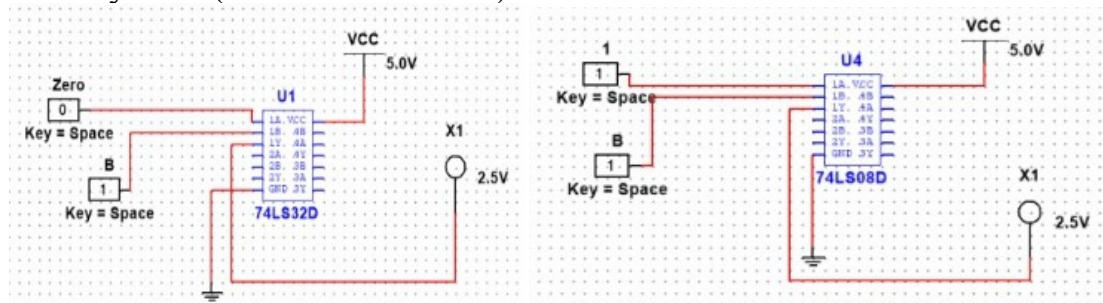


Fig: Simulations of Identity Law

Idempotent Law:

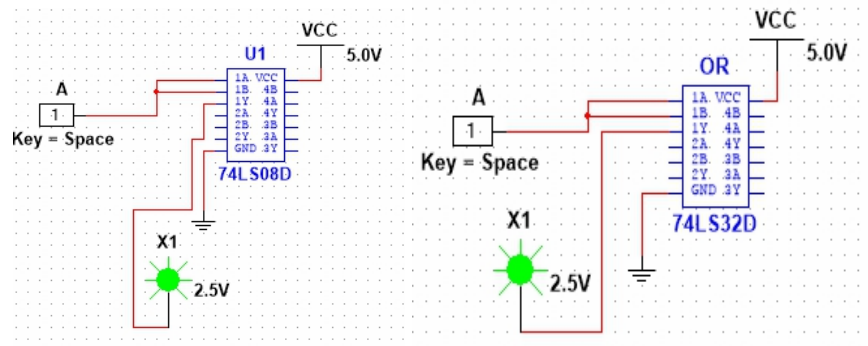
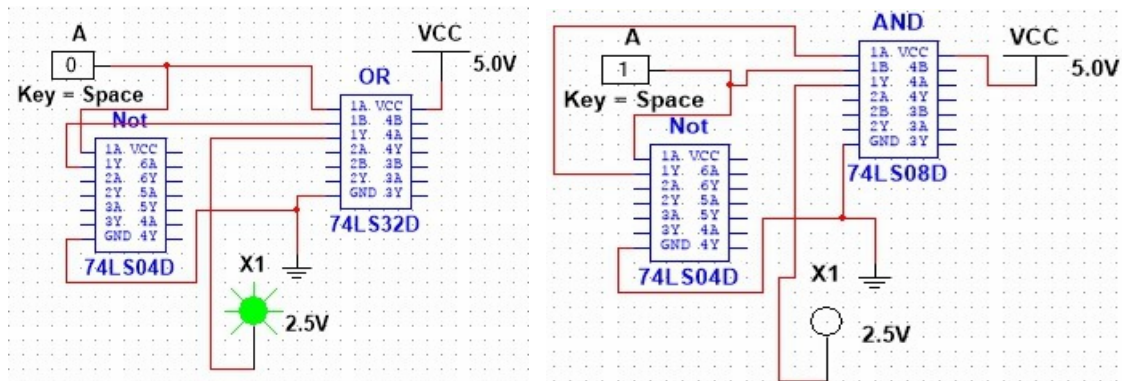


Fig: Simulations of Idempotent Law



Inverse Law:

Fig: Simulations of Inverse Law

Double Complement
Law:

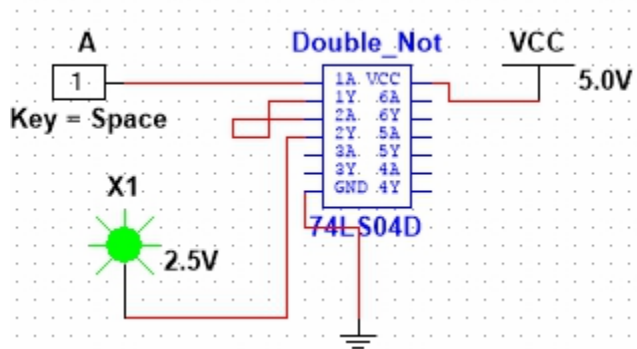


Fig: Simulations of Double Complement Law

Associative Property:

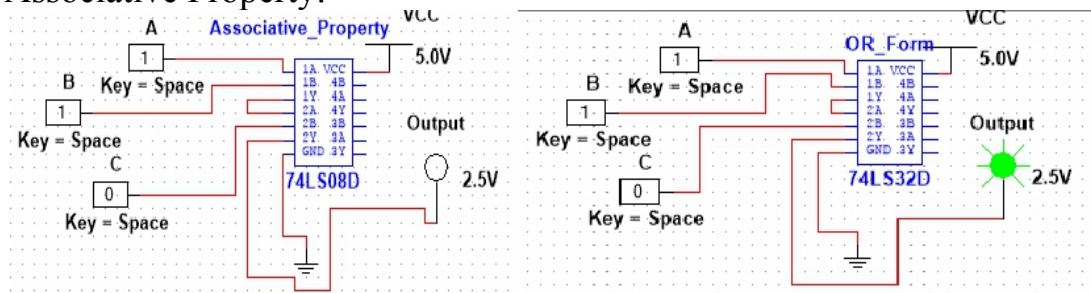


Fig: Simulations of Associative Property

Distributive Property:

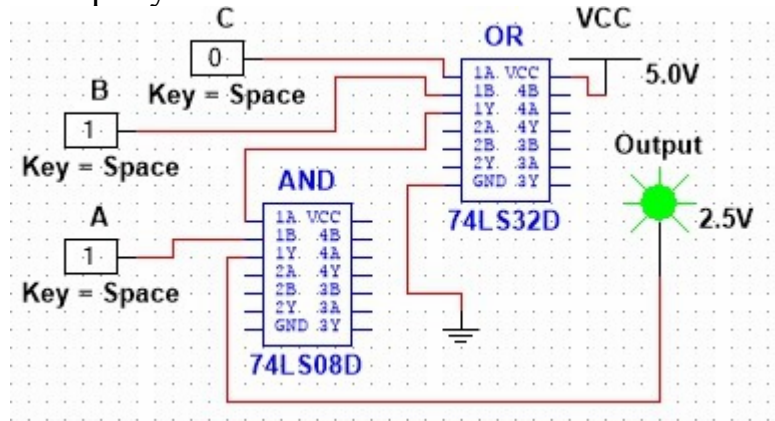


Fig: Simulations of Distributive Property

Observations Calculations: AND Form laws verification table:

A	Identity Law	Null (or Dominance) Law	Idempotent Law	Inverse Law	Double Complement Law
1	1	0	1	0	1
0	0	0	0	0	0

A	Identity Law	Null (or Dominance) Law	Idempotent Law	Inverse Law	Double Complement Law
0	0	1	0	1	0
1	1	1	1	1	1

Boolean properties verification table:

A	B	C	Associative Property (w.r.t +)	Associative Property (w.r.t x)	Distributive Property
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	0	1
1	1	1	1	1	1

